

Background

Dispersion Theory (Liljencrants & Lindblom 1972):

- Vowel spaces tend to maximize psychoacoustic distance between vowel categories
- Optimal n -vowel system minimizes energy function $E = \sum_{i=1}^{n-1} \sum_{j=i+1}^n \frac{1}{d_{i,j}^2}$ ($d_{i,j}$ is pairwise vowel distance)

Becker-Kristal (2010) – large acoustic typology of vowel systems analyzing data from 320 earlier studies:

- Support for broad predictions of DT pooling across languages, but large variation in individual languages
- Even spacing between vowels in F1 in log(Hz) scale, but not psychoacoustic scales like Bark

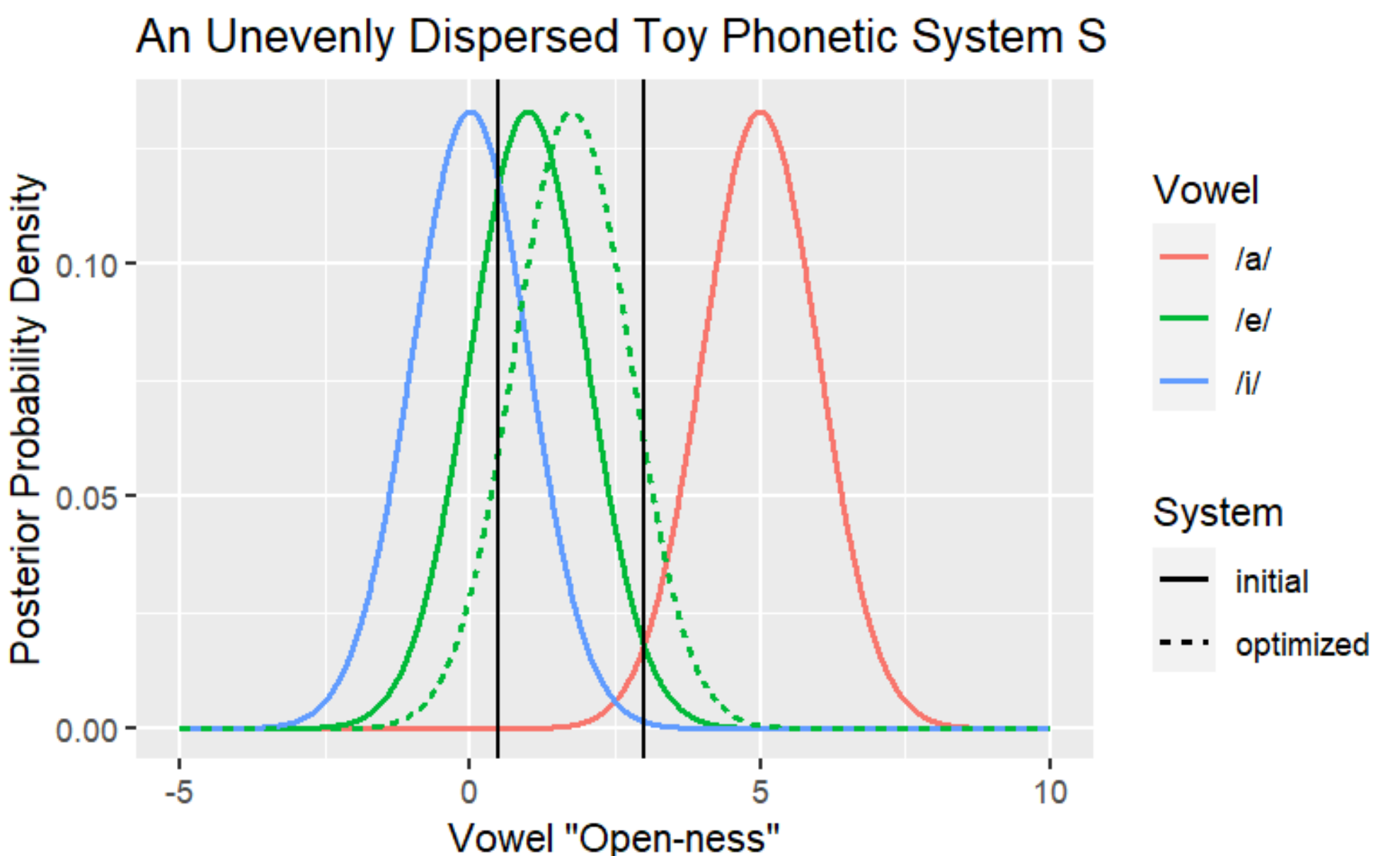
Goal: Develop model of dispersion better able to explain typological facts by modeling interaction of optimal perceptual and articulatory behavior

A New Dispersion Algorithm

Iterated Confusion Minimization (ICM):

- Optimize perceptual classifier on preexisting set of phonetic distributions
- Shift distributions to optimize confusion rate for each category (within articulatorily possible limits)
- Repeat Steps 1-3

An Unevenly Dispersed Toy Phonetic System S



One iteration. Centering /e/ between its perceptual boundaries (black vertical lines) minimizes confusion. Even spacing emerges within 10 iterations.

Acoustic Scales and Even Spacing in F1

Optimal Bayesian classifier boundary b between 1-D Gaussians distributions $N(\mu_1, \sigma^2)$ and $N(\mu_2, \sigma^2)$ with equal variance and prior probabilities p_1 and p_2 :

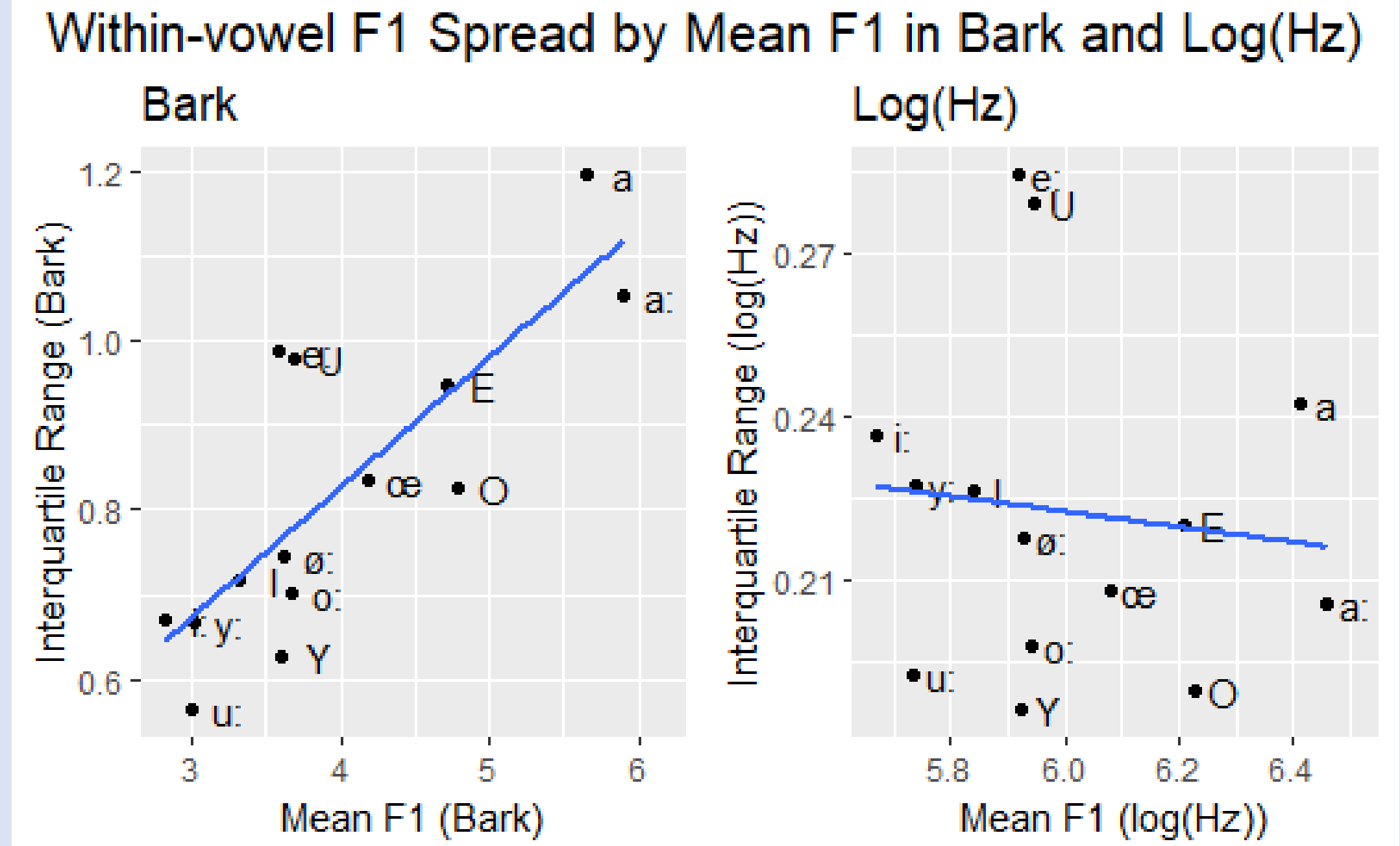
$$b = \frac{\mu_2 + \mu_1}{2} + \frac{\sigma^2 \ln(p_1/p_2)}{\mu_2 - \mu_1}$$

Even spacing only in acoustic scales in which distributions have comparable variance (and if $p_1 = p_2$ for all vowels)

F1 spread increases with mean F1 in psychoacoustic scales, but not log(Hz), cf. German data (Pätzold & Simpson 1997).

- => principled explanation for Becker-Kristal (2010)'s finding of even spacing in log(Hz), not Bark or Mel

Within-vowel F1 Spread by Mean F1 in Bark and Log(Hz)



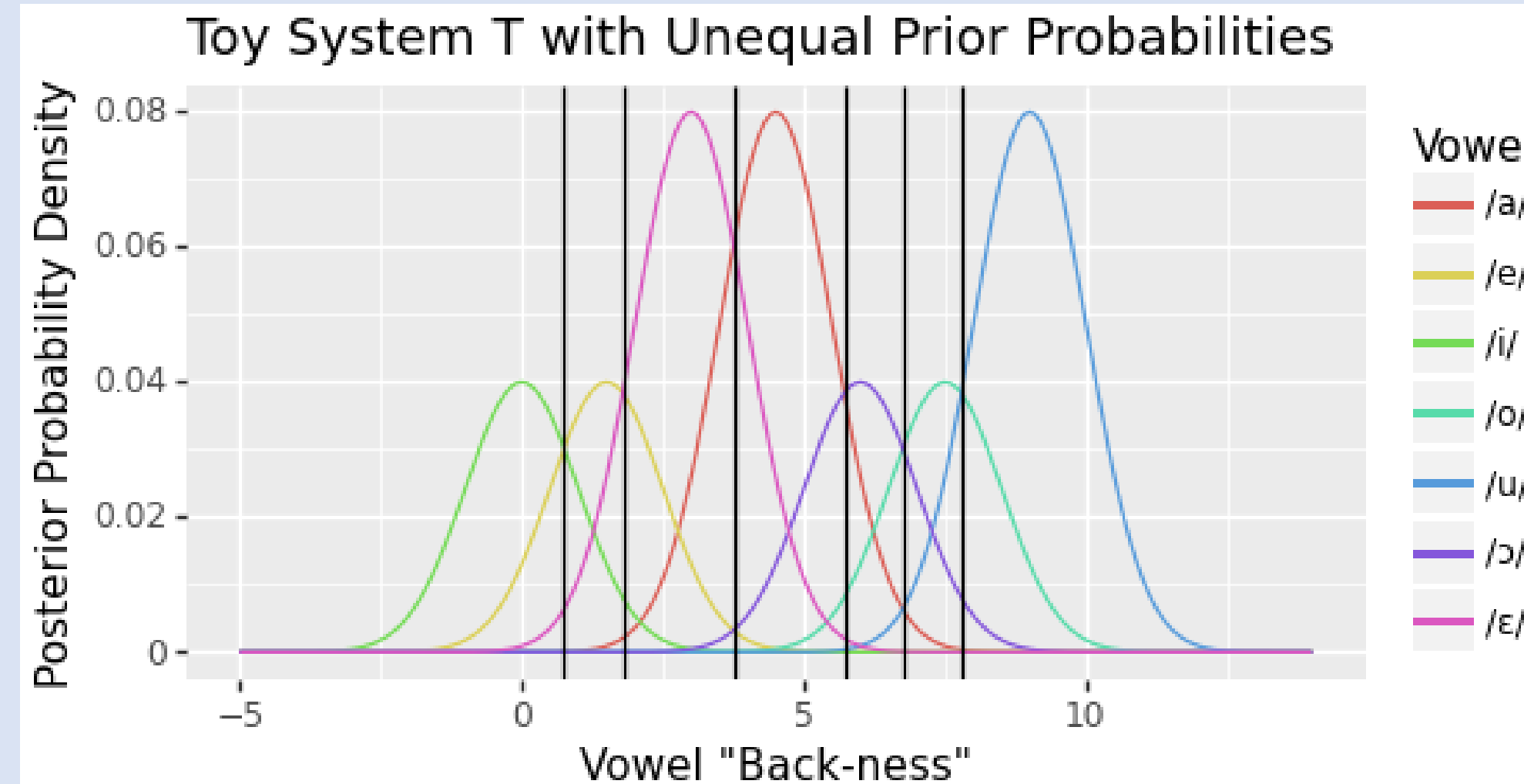
ICM and Unequal Prior Probabilities

Under ICM, phonemes with higher probability exert greater repulsion on others

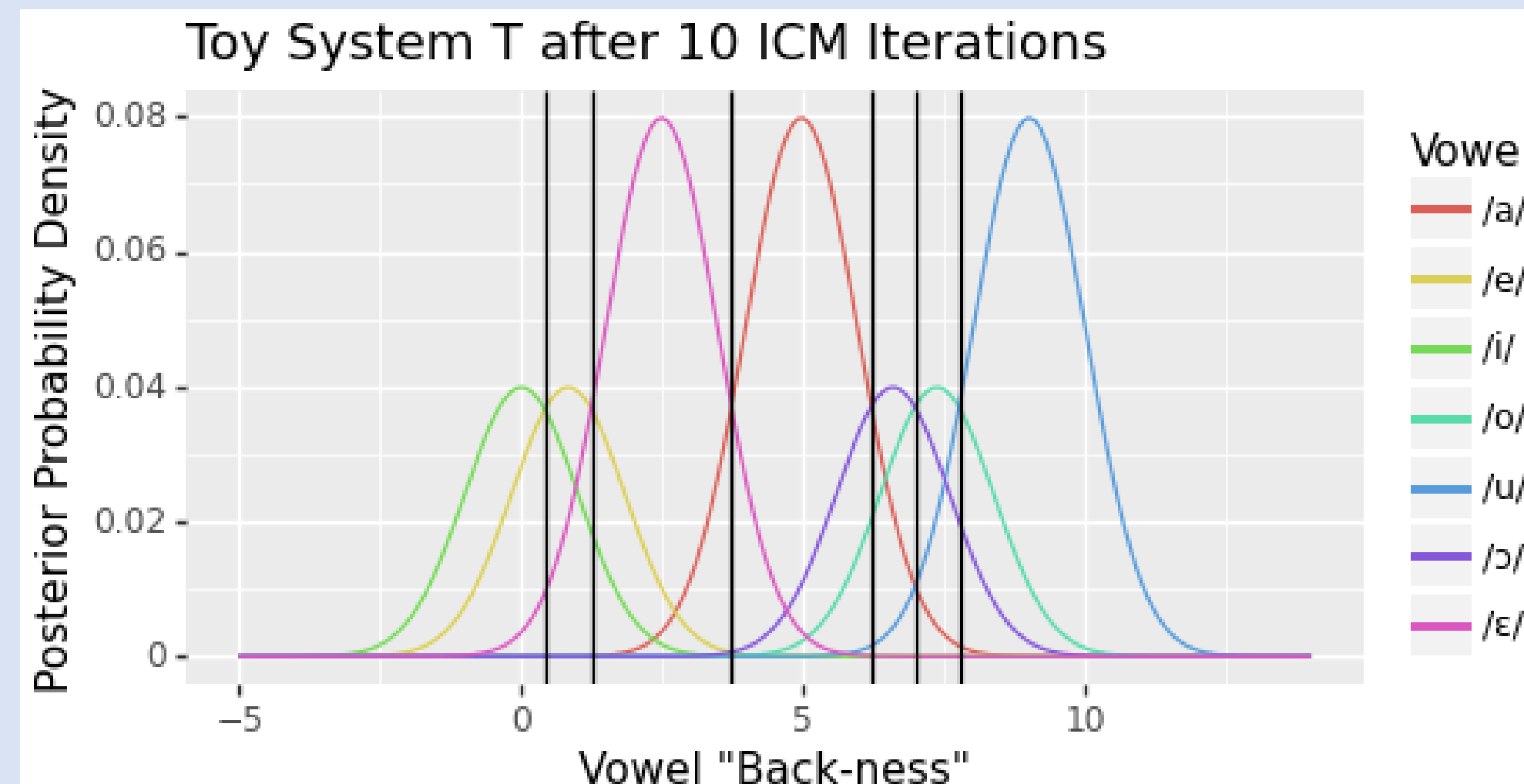
- => clustering of low-probability phonemes

Below: shift away from evenly spaced system that would be stable with equal prior probabilities

Toy System T with Unequal Prior Probabilities

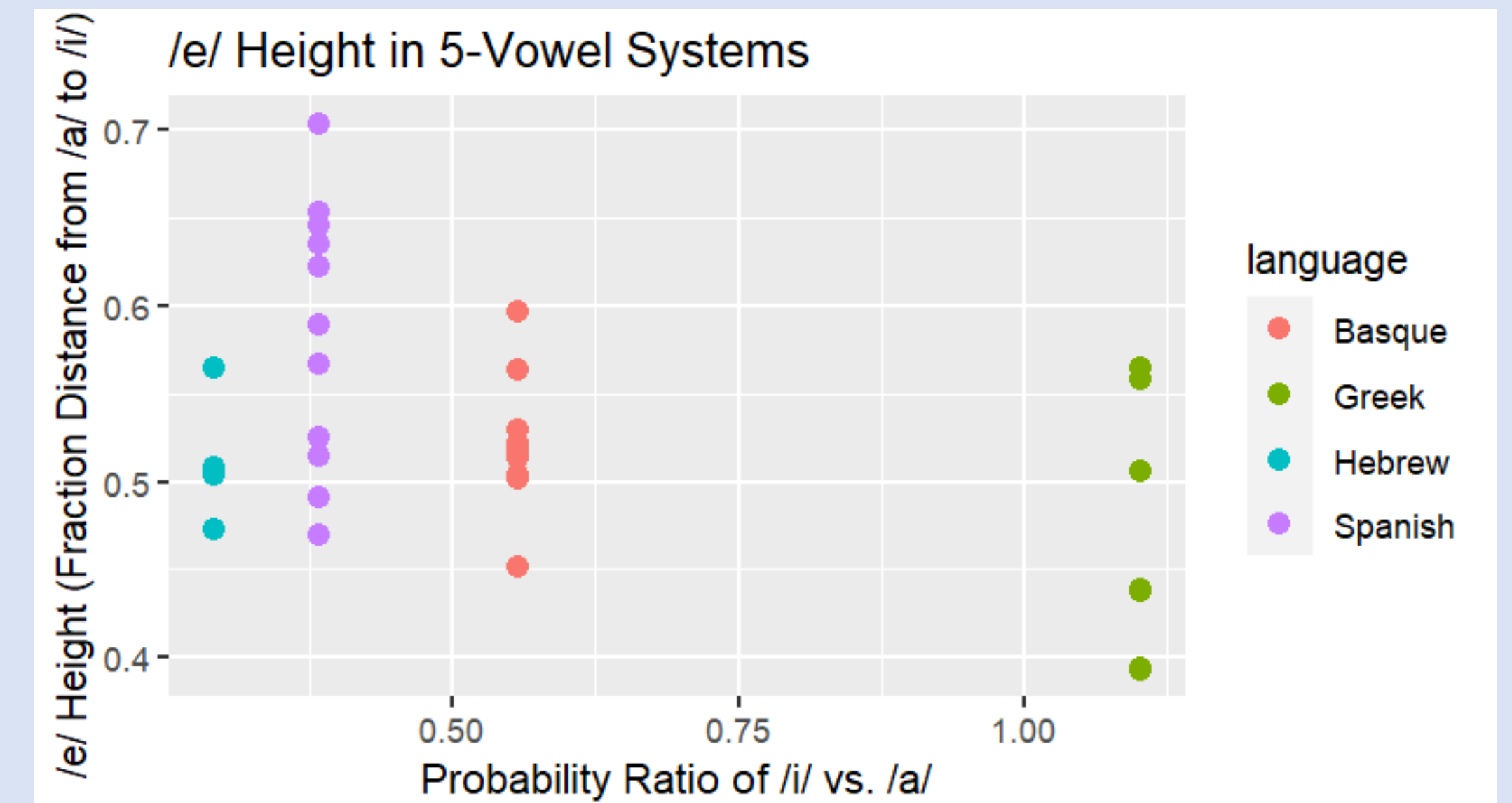


Toy System T after 10 ICM Iterations

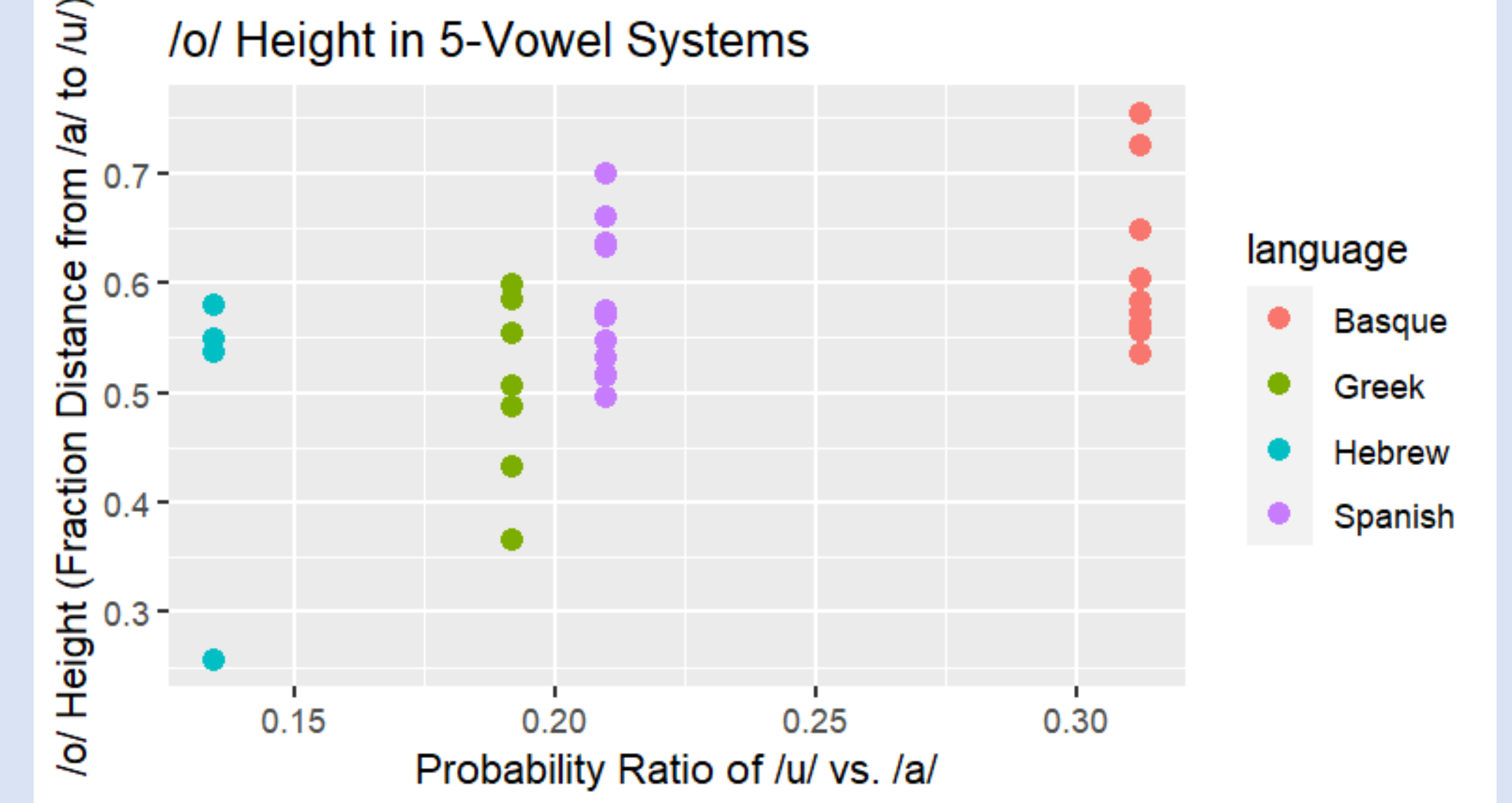


Preliminary look at 5-vowel systems using token frequency for prior probability shows effects ($p < 0.01$), in expected direction for /e/, but in opposite of expected direction for /o/, perhaps due to low sample size.

/e/ Height in 5-Vowel Systems



/o/ Height in 5-Vowel Systems



Discussion

Similar predictions to conventional DT models, with additional principled explanation of empirical facts

Greater potential for explanation of language-specific deviations from “optimal” systems.

- More work required to test if correct predictions made

Incremental change in simulations reminiscent of incremental sound change in real world

- => Potential for modeling of dispersion-related sound changes (e.g. chain-shifts)

Potential Future Work

Testing ICM predictions with more sophisticated operationalizing of prior probabilities

2-D ICM simulations for various languages for direct comparison with Dispersion-Theoretic models

ICM-based modeling of trajectories of historic and ongoing sound changes

Investigate relationship between clustering effects and phonological merger

References

Becker-Kristal, R. (2010). *Acoustic typology of vowel inventories and Dispersion Theory: Insights from a large cross-linguistic corpus*. University of California, Los Angeles

Liljencrants, J., & Lindblom, B. (1972). Numerical simulation of vowel quality systems: The role of perceptual contrast. *Language*, 839-862.

Pätzold, M., & Simpson, A. P. (1997). Acoustic analysis of German vowels in the Kiel Corpus of Read Speech. *Arbeitsberichte des Instituts für Phonetik und digitale Sprachverarbeitung Universität Kiel*, 32, 215-247.

Acknowledgements

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